

Permutations

Definition: A **permutation** of a set S is an ordered sequence that contains every element of S exactly once.

For example, here are all six permutations of the set $\{a, b, c\}$:

(a, b, c) (a, c, b) (b, a, c)

(b, c, a) (c, a, b) (c, b, a)

Ordering is important here.

You can think of a permutation as a ranking of the elements.

How many permutations of an n -element set are there?

So the above question is asking how many rankings of an n -element set.

How many permutations of an n-element set are there?

- There are n choices for the first element.
- For each of these, there are n - 1 remaining choices for the second element.
- For every combination of the first two elements, there are n - 2 ways to choose the third element, and so forth.
- Thus, there are a total of

$$n \cdot (n - 1) \cdot (n - 2) \cdots 3 \cdot 2 \cdot 1 = n! \leftarrow \text{This is called } n \text{ factorial.}$$

permutations of an n-element set.

Example: Permutation

Suppose each digit is an element in $\{1,2,3,4,5,6,7,8,9\}$.

How many 9-digit numbers are there where each nonzero digit appears once?

Each such number corresponds to a permutation of 123456789,
and each permutation corresponds to such a number.

So the numbers of such numbers is equal to

the number of permutations of $\{1,2,3,4,5,6,7,8,9\}$.

Hence there are exactly $9!$ such numbers.

r -permutation

An ordered arrangement of r elements of a set is called an **r -permutation**.

EXAMPLE: Let $S = \{1, 2, 3\}$.

The ordered arrangement 3, 1, 2 is a permutation of S .

The ordered arrangement 3, 2 is a 2-permutation of S .

The number of r -permutations of a set with n elements is denoted by $P(n, r)$.

THEOREM: If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are

$$P(n, r) = n(n - 1)(n - 2) \cdots (n - r + 1)$$

r -permutations of a set with n distinct elements.

COROLLARY: If n and r are integers with $0 \leq r \leq n$, then

$$P(n, r) = n!/(n - r)! .$$

EXAMPLE: How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?

Solution: Because it matters which person wins which prize, the number of ways to pick the three prize winners is the number of ordered selections of three elements from a set of 100 elements, that is, the number of 3-permutations of a set of 100 elements.

Consequently, the answer is

$$P(100, 3) = 100 \cdot 99 \cdot 98 = 970,200.$$

EXAMPLE: How many permutations of the letters *ABCDEFGH* contain the string *ABC* ?

Solution: Because the letters *ABC* must occur as a block, we can find the answer by finding the number of permutations of six objects, namely, the block *ABC* and the individual letters *D*, *E*, *F*, *G*, and *H*.

Because these six objects can occur in any order, there are $6! = 720$ permutations of the letters *ABCDEFGH* in which *ABC* occurs as a block.

Example: In how many ways can 8 papers in an examination be arranged so that the two mathematics papers are not consecutive.

Example: A salesman at a computer store would like to display 5 models of personal computers, 4 models of computer monitors and 3 models of keyboards. In how many ways he arrange them in a row if the items of the same kind are together?

Permutation with repetitions

Suppose a given collection of n objects consists n_1 like objects (say $a, a, a..$), n_2 like objects ($b, b, b...$) and so on n_r like objects ($l, l, l, ...$). Then the total number of permutations of the n things is

$$= \frac{n!}{(n_1! n_2! \dots n_r!)}$$

Example: Find the number of permutations of the letters of the word MATHEMATICS. Also find the number of arrangements beginning and ending with the same letter.

Example. How many positive integers n can be formed using the digits 3, 4, 4, 5, 5, 6, 7, if n has to exceed 50,00,000?

Combinations

How many subsets of size k of an n -element set?

Consider the set $\{1,2,3,4,5\}$ where $n=5$. If $k=2$,

If $k=2$, then there are 10 possible subsets of size 2,

i.e. $\{1,2\}, \{1,3\}, \{1,4\}, \{1,5\}, \{2,3\}, \{2,4\}, \{2,5\}, \{3,4\}, \{3,5\}, \{4,5\}$.

If $k=3$

If $k=3$, then there are also 10 possible subsets of size 3,

i.e. $\{1,2,3\}, \{1,2,4\}, \{1,2,5\}, \{1,3,4\}, \{1,3,5\}$

$\{1,4,5\}, \{2,3,4\}, \{2,3,5\}, \{2,4,5\}, \{3,4,5\}$

Ordering is NOT important here.

Think of it as the number of ways to form a committee of k people from n people.

Combinations

How many subsets of size k of an n -element set?

- There are n choices for the first element.
- For each of these, there are $n - 1$ remaining choices for the second element.
- There are $n - k + 1$ remaining choices for the last element.
- Thus, there are a total of
$$n \cdot (n - 1) \cdot (n - 2) \cdots (n - k + 1)$$
to choose k elements.
when the ordering is important.

So far we counted the number of ways to choose k elements,
when the ordering is important.

e.g. $\{1,2,3\}, \{1,3,2\}, \{2,1,3\}, \{2,3,1\}, \{3,1,2\}, \{3,2,1\}$ will be counted as 6 different ways.

Combinations

How many subsets of size k of an n -element set?

How many different ordering of k elements are (over)-counted?

e.g. If we are forming subsets of size 3, then

$(1,2,3), (1,3,2), (2,1,3), (2,3,1), (3,1,2), (3,2,1)$

are counted as 6 different ways if the ordering is important.

In general, each subset of size k has $k!$ different orderings, and so each subset is counted $k!$ times in the above way of choosing k elements.

• So, when the ordering is not important, the answer is:

This is the
shorthand for
"n choose k"

$$\binom{n}{k} = \frac{n \cdot (n-1) \cdots (n-k+1)}{k!} = \frac{n!}{(n-k)!k!}$$

Example: Team Formation

There are m boys and n girls.

How many ways are there to form a team with 3 boys and 3 girls?

There are $\binom{m}{3}$ choices of 3 boys and $\binom{n}{3}$ choices for 3 girls.

So by the product rule there are $\binom{m}{3} \binom{n}{3}$ choices of such a team.

If $m < 3$ or $n < 3$, then the answer should be zero.

Example: Bit Strings with k Zeros

How many n -bit sequences contain k zeros and $(n - k)$ ones?

We can think of this problem as choosing k positions (out of the n possible positions) and set them to zeroes and set the remaining positions to ones.

So the above question is asking the number of possible positions of the k zeros, and the answer is:

$$\binom{n}{k}$$

Example: Unbalanced Bit Strings

We say a bit string is unbalanced if there are more ones than zeroes or more zeros than ones.

How many n -bit strings are unbalanced?

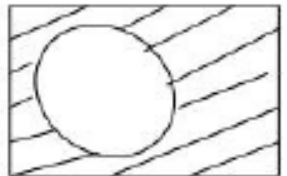
If n is odd, then every n -bit string is unbalanced, and the answer is 2^n .

If n is even, then the number of balanced strings is

$\binom{n}{n/2}$ by choosing $n/2$ positions to zeroes.

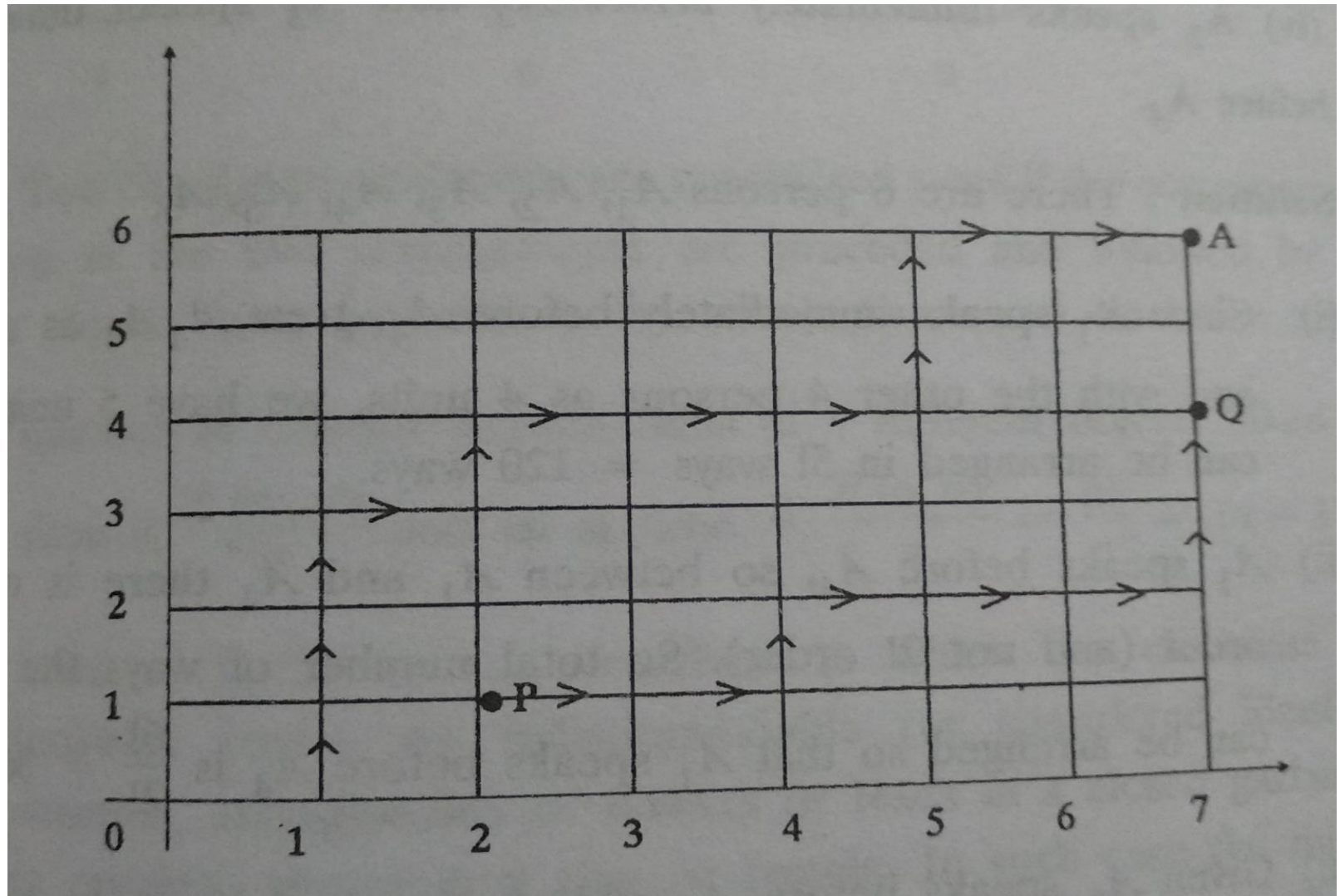
So the number of unbalanced n-bit strings is equal to
the number of all n-bit strings minus the number of balanced strings,
and so the answer is $2^n - \binom{n}{n/2}$

(counting the complement)



Example: Grid problem

- (1) Find the number of paths from 1 to A if one can move along the grid taking steps only right or up at each junction.
- (2) Find the number of paths from the point (2, 1) to (7, 4).



Example. A candidate is required to answer 7 questions out of 12 questions which are divided into two groups, each containing 6 questions. He is not permitted to attempt more than 5 questions from either group. In how many ways can he choose the 7 questions?